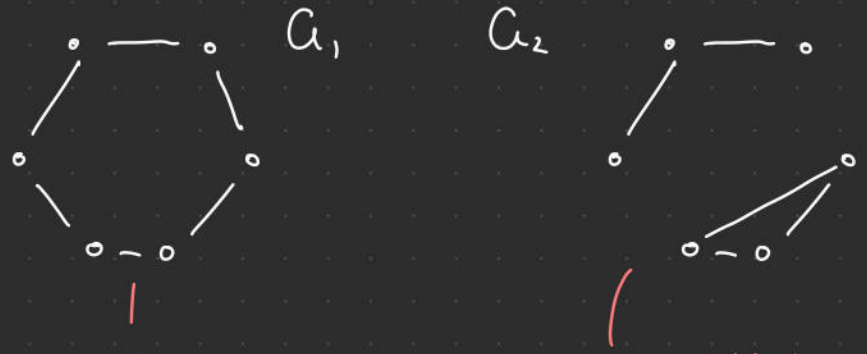
❀ Analytical Graph Theory

- Ramsey theory
- Extremal graphs
- Random graphs
- The Regularity lemma

Ramsey Theory

The party problem:

Six people are at a party. Show that there are 3 people who know eachother or 3 people who are all strangers.



3 people strangers

multiple sets of strangers, 1 set of 3 friends

A clique in a graph G is a connected subgraph of G.

A stable set is a subset which are disconnected

Theorem 3.1 : Every simple graph on 6 vertices contains either a clique or a stable set of size 3.

proof:

Let $G = (V, E)$ be a simple graph on 6 vertices. Let $v \in V$. Then either $\deg(v) < 3$ or $\deg(v) \geq 3$. Assume WOLG that $\deg(v) \geq 3$ (otherwise consider $\bar{G}$)



Let x, y, z be 3 neighbours of v. If any two of the vertices are adjacent, then they form a K_3 subgraph with v. Alternatively, if x, y, z are mutually non-adjacent, then they form a stable set of size 3.

Theorem 3.3: (Ramsey's theorem) 1930

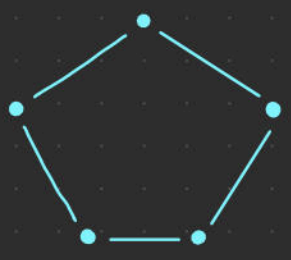
For positive integers s and t and for sufficiently large n, every simple graph on n vertices contains either a clique of size s or stable set of size t.

Ramsey numbers: The Ramsey number $R(s,t)$ is the minimum positive integer n s.t every simple graph on n vertices contains a clique of size s or stable set of size t.

Example 3.2

$1 \leq R(3,3) = 6$?

C_5



C_5 contains neither a clique of size 3, nor a stabel set of size 3.

Hence, together with theorem 3.3, this proves that $R(3,3) = 6$.

Proposition 3.5: $R(s,1) = 1$ for all $s \geq 1$.

Proof:

Let G be a graph on $n \geq 1$ vertices:

G  Then G always contains a stable set of size 1

Proposition 3.6: $R(s,2) = s$ for all $s \geq 1$.

"either the graph already contains a clique of size s or two of its vertices are disjoint"

proof:

This statement is trivially true for K_1, so assume that $s \geq 2$. This statement is true for $G = K_s$ (since it contains a clique of size s). So now assume that G is not complete, then there exists 2 vertices which are not adjacent and hence form a stable set of size 2.

(so we have that $R(s,2) \leq 2$, it remains to show that $R(s,2) \not\leq 2$.)

Consider the graph G' which has at most $s-1$ vertices. If $G' = K_{s-1}$, it contains neither a clique of size s nor a stable set of size 2. Which means that $R(s,2) > s-1$. Hence $R(s,2) = s$ for all $s \geq 1$.

There are no more general results.

Theorem 3.7: (Rameys theorem restated)

For positive integers s and t, there exists a (minimal) number $R(s,t)$ such that if we colour the edges

of $K_{R(s,t)}$ red and blue, this graph will either have a K_s subgraph with only red edges or a K_t subgraph with only blue edges.

Proof:

We use induction on s and t. The induction will proceed as follows:

We first prove the base cases:
$R(s,1), R(s,2), R(1,t), R(2,t)$ exist for s and t.

Then we prove the induction step:
If $R(s,t-1)$ exists and $R(s-1,t)$ exists, then $R(s,t)$ exists.

By proposition 3.5 and 3.6, we have that

$$R(s,1) = 1,\ R(1,t) = 1,\ R(s,2) = s,\ R(2,t) = t.$$

So the statement holds for $s+t = 3$.

Now we assume that the statement is true for $s+t-1 \geq 2$, i.e. $R(s-1,t)$ and $R(s,t-1)$ exist for $s,t \geq 2$.

Consider a 2-edge-coloured complete graph G on $R(s,t-1) + R(s-1,t)$ vertices. Each edge of G is coloured either red or blue. Let v be a vertex of G. Since G is a complete graph, v has $R(s,t-1) + R(s-1,t)$ neighbours.



Let N be the number of red edges incident to v
Let M " " " blue " " "

It follows that

$$M + N + 1 = R(s, t-1) + R(s-1, t).$$

This means that

$M \geq R(s, t-1)$ or

$N \geq R(s-1, t)$.

If $M \geq R(s, t-1)$ then the induced subgraph on M vertices in G has a red K_s or blue K_{t-1}. If it is the former, we are done. If it is the latter, then we observe that we can form a blue K_t by adding the vertex v to K_{t-1} along with the M blue edges and we are done.

If $N \geq R(s-1, t)$, then the induced subgraph on those N vertices in G has either a red K_{s-1} or a blue K_t. Then the argument follows that above.

Hence $R(s, t) \leq R(s, t-1) + R(s-1, t)$ and the result follows.

Corollary 3.8: For all $s, t \geq 2$, $R(s, t) \leq R(s-1, t) + R(s, t-1)$. Moreover, if $R(s-1, t)$ and $R(s, t-1)$ are both even, then the inequality is strict.

Proposition 3.9: $R(3, 4) = 9$.

From theorem 3.1 and ex. 3.2, we have that $R(3, 3) = 6$. From proposition 3.6 we have that $R(2, 4) = 4$. Since these are both even, applying corollary 3.8, we have

$$R(3, 4) < R(2, 4) + R(3, 3) = 4 + 6 = 10.$$

Hence we have that $R(3,4) \leq 9$. It remains to be shown that $R(3,4) \neq 8$.

Proposition 3.10: $R(3,5) = 14$

(proof like that of prop 3.9)

Proposition 3.11: $R(4,4) = 18$

Proposition 3.12: $R(4,5) = 25$

Conjecture: $R(5,5) = 45$

Proposition 3.13: $43 \leq R(5,5) \leq 48$

Theorem 3.14: $R(s,t) \leq \binom{s+t+2}{s-1}$

Corollary 3.15: $R(s,t) \leq 2^{s+t-2}$

Theorem 3.16: $R(s,s) \geq 2^{s/2}$

proof:

Assume $s \geq 3$. Let $\mathcal{G}_n$ denote the set of simple graphs with vertex set $\{v_1, \ldots, v_n\}$. Let $\mathcal{G}_n^s$ be the set of these labelled graphs which have a clique of order s.

Observe that $|\mathcal{G}_n| = 2^{\binom{n}{2}}$ ⊛.

Since each subset of the $\binom{n}{2}$ possible edges $v_i v_j$ $(1 \leq i \leq j \leq n)$ determines a graph in $\mathcal{G}_n$.

Similarly, the number of graphs in $\mathcal{G}_n$ which have a particular set of s vertices in a clique of order s is $2^{\binom{n}{2}-\binom{s}{2}}$ (there are no choices for these edges).

Because there are $\binom{n}{s}$ places in a graph in which a K_s subgraph can appear,

we have: $|\mathcal{G}_n^s| \leq \binom{n}{s} 2^{\binom{n}{2}-\binom{n}{s}}$ ☺ .

The inequality arises because graphs in $\mathcal{G}_n^s$ which have more than one K_s subgraph are counted more than once by the expression on the right-hand side of ☺ .

Hence, by ⊛ and ☺ we have ★ ★ ★

$$\frac{|\mathcal{G}_n^s|}{|\mathcal{G}_n|} \leq \frac{\binom{n}{s} 2^{\binom{n}{2}-\binom{n}{s}}}{2^{\binom{n}{2}}}$$

$$= \binom{n}{s} 2^{-\binom{s}{2}}$$

$$< \frac{n^s 2^{-\binom{s}{2}}}{s!}$$

$$\binom{n}{k} = \frac{n!}{k!(n-k)!} < \frac{n^k}{k!}$$

Now suppose that $n < 2^{s/2}$. Then

$$\frac{|\mathcal{G}_n^s|}{|\mathcal{G}_n|} < \frac{(2^{s/2})^s 2^{-\binom{s}{2}}}{s}$$

$$= 2^{s^2/2 - \binom{s}{2}}$$

$$= \frac{2^{s/2}}{s!}$$

$$< \frac{1}{2}.$$

In other words, if $n < 2^{s/2}$, then fewer than half of the graphs in $\mathcal{G}_n$ contain a clique of s vertices. Likewise, by complementarity, fewer than half of the graphs in $\mathcal{G}_n$ contain a stable set of s vertices.

Hence, there exists at least one graph in $\mathcal{G}_n$ that contains

neither a clique or a stable set of s vertices. Because this holds for all $n < 2^{s/2}$, we have the Ramsey number must be $\geq 2^{s/2}$.

2. Extremal Graphs (Turán's Theorem)

Extremal graph theory is primarily concerned with determining the maximum number of edges that a simple graph of order n may contain without having a given property.

Theorem 3.18 (Mantel's thm) 1907.

If G is a graph of order $n \geq 3$ and size $m > \lfloor \frac{n^2}{4} \rfloor$, then G contains a triangle.

proof:

Suppose the thm is false. Then there exists a smallest integer n for which the statement is false.

So there is some graph $G = (V, E)$ of order n and size $\lfloor \frac{n^2}{4} \rfloor + 1$ that contains no triangle. Let $u, v \in V$ and $uv \in E$ and let $v_1, \ldots, v_{n-2}$ be the remaining vertices of G.



Since G contains no triangles, then at most one of u and v is adjacent to v_i $(1 \leq i \leq n-2)$. This implies that $\deg(u) + \deg(v) \leq n$.

Let $H = G - \{v\} - \{u\}$. Then H has order $(n-2)$ and size m where

$$m = \left\lfloor \frac{n^2}{4} \right\rfloor + 1 - (\deg(u) + \deg(v) - 1),$$

and so

$$m > \left\lfloor \frac{n^2}{4} \right\rfloor - n + 2$$

$$= \left\lfloor \frac{(n-2)^2}{4} \right\rfloor + 1 ,$$

since H has fewer vertices than G and so by assumption it must contain a triangle. This is a contradiction.

If n is even in thm 3.18, say $n = 2k$, then every graph of size $m > k^2$ contains a triangle.

This bound cannot be improved because there exists a graph of order $2k$ and size k^2 that contains no triangles.

That graph is the complete bipartite graph $K_{k,k}$, which has $2k$ vertices and k^2 edges.

In fact, this is the only such graph.

Theorem 3.19: Let G be a graph of order $n \geq 3$ and size m. If $m > \left\lfloor \frac{n^2}{4} \right\rfloor$ or $m = \left\lfloor \frac{n^2}{4} \right\rfloor$ and $G \neq K_{\lfloor \frac{n}{2} \rfloor, \lfloor \frac{n}{2} \rfloor}$, then G contains a triangle.

Let F be a simple graph, we denote $ex(n, F)$ the maximum number of edges in a graph G on n vertices (number) which <u>does not</u> contain a copy of F.

Such a graph is called an extremal graph (for this property) and the set of extremal graphs is denoted $Ex(n, F)$. (set)

Exercise 3.4

We have $ex(n, K_3) = \left\lfloor \frac{n^2}{4} \right\rfloor$ and $Ex(n, K_3) = \left\{ K_{\lfloor \frac{n}{2} \rfloor, \lceil \frac{n}{2} \rceil} \right\}$

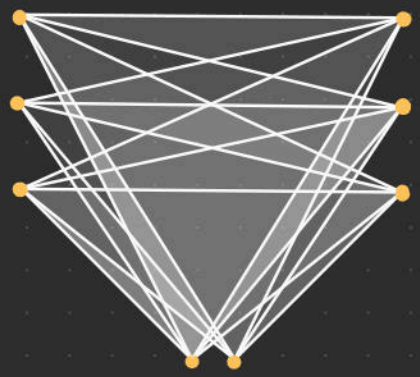
$\Rightarrow$ there is only one extremal graph for $ex(n, K_3)$.

The graph $K_{\lfloor \frac{n}{2} \rfloor, \lceil \frac{n}{2} \rceil}$ is an example of a **Turán Graph**.

The Turán graph, denoted $T_{n,r}$ or $T(n,r)$, is a complete r-partite graph on $n \geq r$ vertices whose partition sets differ in size by at most 1.

In general: $K_{\lfloor \frac{n}{r} \rfloor, \lceil \frac{n}{r} \rceil \cdots \lceil \frac{n}{r} \rceil, \lceil \frac{n}{r} \rceil}$.

Exersise 3.5: Turán graph $T(8,3)$



$K_{2,3,3}$

* Does not contain K_4
* Extremal for K_4
* r-partitions wont contain K_{r+1}

In fact, $Ex(8, K_4) = \{T(8,4)\}$

Theorem 3.20: Among all bipartite graphs of order n, the Turán graph $T(n,r)$ is the unique graph of maximum size. Turán graphs are dense, they have about n^2 edges. Moreover, let $t(n,r)$ denote the number of edges in $T(n,r)$, then

$$t(n,r) \leq \frac{n^2}{r}\left(1 - \frac{1}{r}\right),$$

with equality when r divides n

Theorem 3.21: If a graph $G = (V, E)$ on n vertices has no $(r+1)$-clique $(1 \leq r \leq n-1)$, then

$$|E| \leq \frac{n^2}{r}\left(1 - \frac{1}{r}\right) \quad \circledast$$

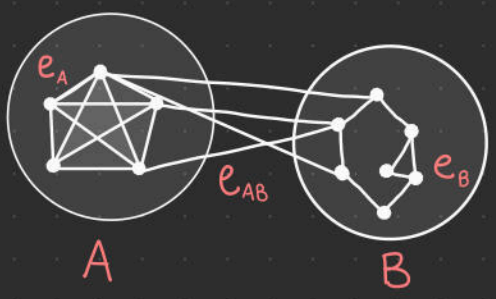
Theorem 3.22: (restating theorem 3.21)
If a graph $G = (V, E)$ on n vertices has no $(r+1)$-clique $(1 \leq r \leq n-1)$ and $ex(n, K_{r+1})$ edges, then $G = T(n, r)$.

Note, for $r = 1$ this is trivial. For $r = 2$, we have $ex(n, k_3) = \left\lfloor \frac{n^2}{4} \right\rfloor$. Mantel's theorem is a 'special case' of Turáns theorem.

Proof of 3.21 (by induction on n)
When $n \leq r$, $\circledast$ trivially holds (G is too small).
Assume $n > r$ and the statement holds for all graphs with r to $n-1$ vertices.

Let G be a graph on vertex set $V = \{v_1, \ldots, v_n\}$, without $(r+1)$-cliques and with a maximum number of edges where $n > r$.

G must have at least one K_r-clique, otherwise it would not have maximum size.



Let A be a K_r subgraph in G and let $B = V \setminus A$. We know that A contains $e_A = \binom{r}{2}$ edges. We now estimate e_B, the number

of edges in B, and e_{AB}, the number of edges between vertices in A and vertices in B. By our induction hypothesis, we have

$$e_B \leq \frac{(n-r)^2}{2}\left(1-\frac{1}{r}\right).$$

Since G has no K_{r+1} subgraph, every vertex $v_i \in B$ ($1 \leq i \leq |B|$) is adjacent to at most r-1 vertices in A. We have

$$e_{AB} \leq (r-1)(n-r).$$

Hence, we have

$$|E| = e_A + e_{AB} + e_B \leq \binom{r}{2} + \frac{(n-r)^2}{2}\left(1-\frac{1}{r}\right) + (r-1)(n-r)$$

$$\leq \frac{n^2}{2}\left(1-\frac{1}{r}\right)$$



Might be content missing here

3. Random Graphs

A random graph is a graph in which the number of vertices is specified and their adjacencies are determined at random (or, in a random way).

We may redard the set of labelled graphs on n vertices as our sample space. The result of selecting one element G of this sample space, according to a probability function p is called a random graph. {a set and probability function}

Erdös and Rengi (the simplest model)

* 2 models / variations
  ~ both originate from 1947 paper.

Let $N = \binom{n}{2}$.

① The uniform model - $\mathcal{G}(n,m)$. {can fix # edges}

For $0 \leq m \leq N$, (all or no edges) the probabitity space $\mathcal{G}(n,m)$ consists of all $\binom{N}{m}$ subgraphs of K_n with m edges and uniform probability function

$$P(G) = \frac{1}{\binom{N}{m}} \quad \text{for all } G \in \mathcal{G}(n,m).$$

② The binomial model - $\mathcal{G}(n,p)$

The probability space $\mathcal{G}(n,p)$ is obtained by fixing a probability p between 0 and 1, and choosing

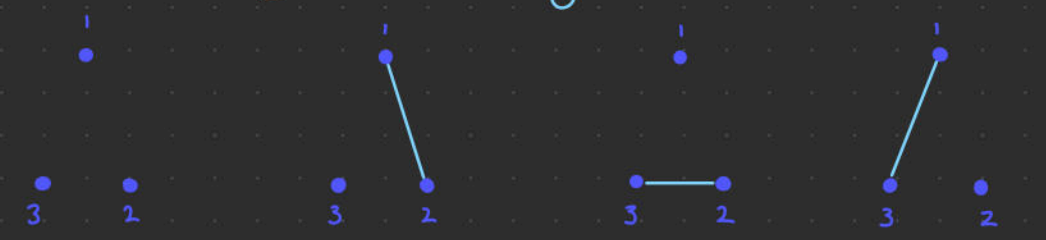
each edge with probability p. {These choices are independent}. Since $(1-p)$ is the probability of an absent edge, we have probability function

$$P(G) = p^m (1-p)^{N-m}, \text{ for each } G \in \mathcal{G}(n,p).$$

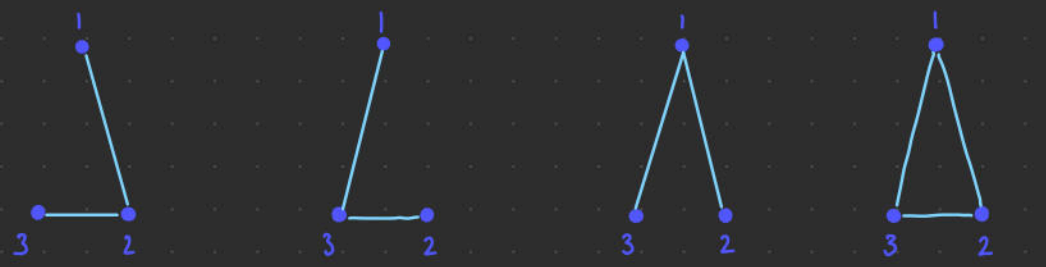
These two models are equivalent when n is large and $p = \dfrac{m}{\binom{n}{2}}$.

Example 3.5

The probability space $\mathcal{G}(3,p)$



$p^0(1-p)^3$ $p^1(1-p)^2$ $p(1-p)^2$ $p(1-p)^2$



$p^2(1-p)$ $p^2(1-p)$ $p^2(1-p)$ $p^3(1-p)^0$

Exercise: play with different values of p

i) $P(G \text{ is } \cdot\ \cdot\ \cdot) = \binom{3}{0}(1-p)^3 = (1-p)^3$

ii) $P(G \text{ is } \cdot\!-\!\cdot\ \cdot) = \binom{3}{1}p(1-p)^2 = 3p(1-p)^2$

iii) $P(G \text{ is } \wedge) = \binom{3}{2}p(1-p) = 3p^2(1-p)$

iv) $P(G \text{ is } \triangle) = \binom{3}{3}p^3 = p^3$

v) $P(G \text{ is connected}) = \binom{3}{2}p^2(1-p) + \binom{3}{3}p^3 = p^2(3-2p)$

Properties of "almost all graphs"

We are interested in the properties of our probability space as $n \to \infty$. We say that a typical element of our space has property Q when the probability that the random graph $G \in \mathcal{G}(n,p)$ on n vertices has Q Q tends to 1 as $n \to \infty$.

We also say that 'almost all graphs' or 'almost every graph has property Q' or 'property Q almost almost always holds'.

Consider $\mathcal{G}(n,p)$. We want to know about $\lim_{n\to\infty} \mathcal{G}(n,p)$

* Is a randomly chosen graph connected?
* Does a " " " contain triangles?
* What is the expected number of edges?
* Is a randomly chosen graph Hamiltonian?

Questions are most interesting (difficult) when $p = c/n$ (c constant).

Lemma 3.22: When p is constant, for every graph H, almost every graph $G \in \mathcal{G}(n,p)$ contains an induced copy of H.

Let $G[u]$ be the subgraph of G induced by the vertex set U.

(Proof)
Let H have k vertices. If $n \geq k$ and $U \subseteq \{0,1,\ldots,n-1\}$ is a fixed set of vertices of G, then $G[u]$ is isomorphic to H with a certain probability $r > 0$. The probability r depends on p (but not on n).



Graph G contains a collection $\lfloor \frac{n}{k} \rfloor$ disjoint such sets U. The probability that none of the corresponding graphs $G[U]$ are isomorphic to H is $(1-r)^{\lfloor \frac{n}{k} \rfloor}$, since these events are independent, as the edge sets of the vertices of G are disjoint. Thus

$$P(G \text{ does not contain } H) \leq (1-r)^{\lfloor \frac{n}{k} \rfloor}.$$

Since $(1-r)^{\lfloor \frac{n}{k} \rfloor} \to 0$ as $n \to \infty$, then the result holds.

Some properties of almost all graphs; p constant.

Theorem 3.23: In almost every graph $G \in \mathcal{G}(n,p)$, every pair of vertices is connected by a path of length 2.



Theorem 3.24: Almost every graph $G \in \mathcal{G}(n,p)$ is connected.

Theorem 3.25: For $k > 0$, almost every graph $G \in \mathcal{G}(n,p)$ is k-connected.

Threshold Functions

For many graph properties, the limiting probability that a random graph possesses them jumps from 0 to 1 (or vice versa) very rapidly. That is, with a small increase in the expected number of edges).

The number of triangles in $G \in \mathcal{G}(n,p)$ changes abruptly at $p = 1/n$.

* If $p << 1/n$, then almost every graph $\in \mathcal{G}(n,p)$ has no triangles.

* If $p >> 1/n$, then a graph $G \in \mathcal{G}(n,p)$ has at least one triangle.

The function $f(n) = 1/n$ is the threshold function for triangles. A threshold function for a graph property Q is a function $f(n)$ s.t

* If $p << f(m)$, almost all $G \in \mathcal{G}(n,p)$ do not have Q.
* If $p >> f(n)$, " " " do have Q.

All monotone graph properties have threshold functions

preserves property as # edges strictly increasing / decreasing

The evolution of random graphs

In a random graph $G \in \mathcal{G}(n,p)$ (where n is arbitrarily large), when we increase p from 0 to 1, it is possible to determine at which stage in the 'evolution' a particular graph property Q will appear.

Rough timeline:

* $p = 0$: G consists of isolated vertices (G is a stable set).
* $p = 1/n^2$: The expected or average number of edges in is 1.
* $p = d/n^2$ $(d > 1)$: " " " is almost surely d, and the components are of size 1 or 2.
* $p = \frac{1}{n^{k/k-1}}$: The components of G are trees.
* $p = 1/n$: cycles are born
* $p = d/n$ $(d > 1)$: There are a constant number of components. Almost surely all components are trees or unicyclic and of size at most $\log n$.
* "Soon after" : G is non-planar
* $p = d/n$ $(d \sim n)$: a large component forms, and it has size $n^{2/3}$.
* $p = d/n$ $(d > 1)$: The largest component has size Cn $(0 < c < 1)$ - the "double jump".
* "Shortly after" : G is almost surely Hamiltonian.

4. The Regularity Lemma

Let $G = (V, E)$ be a graph and let $X, Y \subseteq V$ be disjoint. Let $e(X,Y)$ be the number of edges between the vertices in X and the vertices in Y. The density of (X,Y) is

$$d(X,Y) = \frac{e(X,Y)}{|X| \cdot |Y|}.$$

Let ε $(0 < \varepsilon \leq 1)$ be a small positive constant. We refer to a subset S of X as small if $|S| \leq \varepsilon |X|$ and large if $|S| \geq \varepsilon |X|$.

Given some $\varepsilon > 0$, we call a pair (X,Y) ε-regular if

$$|d(X',Y') - d(X,Y)| \leq \varepsilon \quad \text{whenever } X' \subseteq X \wedge Y' \subseteq Y,$$

and "large subset"

$$|X'| \geq \varepsilon |X| \quad \wedge \quad |Y'| \geq \varepsilon |Y|.$$

This tells us that the edges between ε-reglar parts are distributed fairly uniformly.

More so for small ε.

Consider a partition $\{V_0, \ldots, V_k\}$ of V in which V_0 singled out as an exceptional set (might be empty). We call such a partion an ε-regular partition if

(i) $|V_0| \leq \varepsilon |V|$ (small subset)

(ii) $|V_1| = |V_2| = \cdots = |V_k|$

(iii) All but at most εk^2 of the pairs (V_i, V_j) with $1 \leq i < j \leq k$, are ε-regular

Theorem 3.26: For every $\varepsilon > 0$ and every integer $m \geq 1$, there exists an integer M such that every graph of at least order m has an ε-regular partition $\{V_0, V_1, \ldots, V_k\}$ with $m \leq k \leq M$.

What this lemma tells us:

* Every sufficiently large graph has an ε-regular partition into not too many parts.
* The upper bound M ensures that for large graphs the parts are large

  → ε-regularity is trivial when parts are singletons and powerful when they are large.

* We can specify a lower bound, m, for the # of part. sets.

Note: Designed to be used with dense graphs. In sparse graphs, it is trivial because all densities of pairs tend to 0.

We can use the Regularity Lemma to prove that a [redacted] edge density of a graph G suffices to force the occurence of some given subgraph H. We do this by using the 'fairly' uniform dis-

tribution of edges between (v_i, v_j) pairs and showing that they confirm the subgraph.

Ramsey Numbers:

Recall that the best bounds we have on $R(s,s)$ are

$$2^{s/2} \leq R(s,s) \leq 4^{s-1}.$$

Define $R(G,G)$ as the Ramsey number of a simple graph G, as the smallest n for which a 2-edge-colouring of K_n yeilds a monochromatic copy of G. (more general than $R(s,t)$) An obvious bound is $R(G,G) \leq R(s,s)$.

Theorem 3.27: For any graph G of maximum degree Δ we have

$$R(G,G) \leq C_n,$$

where C depends only on Δ.

The regularity lemma can be used to prove that the Ramsey number of graphs of most degree Δ grow linearly with [redacted] order.

Corollary 3.28: For any graph G of max degree Δ, $R(s,s) \leq C_n$ (a better bound).

Theorem 3.29: (Erdös-Stone thm)
For all integers $r, s \geq 1$ and every $\varepsilon > 0$, there exists an integer n_0 s.t every graph with $n \geq n_0$ vertices and at least

$$t(n,r) + \varepsilon n^2$$

edges, contains a K_{r+1} subgraph.

Note: K_{r+1}^{s} means that we have s copies of K_{r+1} subgraphs.